ARVIN F. BAÑES

Computer Science Student | Web/Mobile Developer

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience designing and building full-stack and mobile applications end-to-end, from architecture to deployment. Proficient in Next.js, Prisma, PostgreSQL, and Flutter, with practical experience integrating AI/ML pipelines (computer vision, depth estimation, LLM APIs) into production-oriented apps. Developed a no-code SaaS platform for Filipino SMBs (booking, e-commerce, and POS modules), an offline-first mobile POS system, and a hyperlocal social platform concept — alongside a thesis project applying AR-anchored monocular depth fusion for food volume estimation. Comfortable working across the stack, from database schema design and payment integrations to AR/computer vision and API architecture.

SKILLS

- **Languages:** Dart, JavaScript/TypeScript, Python, SQL
- **Frontend/Mobile:** Next.js (App Router), Flutter, Tailwind CSS
- **Backend:** Prisma ORM, FastAPI, NextAuth (Google OAuth)
- **Databases:** PostgreSQL (Neon), SQLite/Isar
- **AI/ML & Computer Vision:** ARKit/ARCore, Depth Anything V2, YOLOv8-Seg, RANSAC, Gemini API, ML Kit (barcode scanning)
- **Payments/Integrations:** PayMongo, Google Maps Platform APIs
- **Tools & Workflow:** Git/GitHub, Google Antigravity, Claude Code
- **Languages:** English, Filipino, Hiligaynon/Ilonggo (bilingual/native proficiency)

EXPERIENCE

Software Engineering Capstone — Peer-to-Peer Online Tutoring System
[2025] – [2026]

- Designed and developed the Smart Peer Tutoring System (SPTS), a web-based platform connecting students for peer tutoring, built with Next.js, Node.js/Express, and SQLite
- Defended the project as a capstone deliverable for the Software Engineering course

Programming 2 — AI-Integrated Note-Taking Web App(*NOTER*)
[2025]

- Built a web app combining note-taking with AI-generated quizzes to reinforce learning from user notes

Independent Project — Online POS System for a Milk Tea shop(*Barista's Milk Tea*)
[Start Year] – Present

- Web based POS system for *Batristas Milktea* shop and cashier realtime supervision automated operations

Undergraduate Researcher — University of San Agustin
[2025] – Present

- Leading thesis research on AR-anchored monocular depth fusion for food volume estimation, integrating computer vision and depth estimation models into a cross-platform mobile app

FEATURED PROJECT

AR Food Volume Estimation Mobile App — Undergraduate Thesis, University of San Agustin*[Start Date] – [Completion Date]*

- Built a cross-platform mobile application (Flutter) that estimates food volume using augmented reality and monocular depth estimation.

- Implemented a computer vision pipeline combining Depth Anything V2 for depth estimation, YOLOv8-Seg for food segmentation, and ARKit/ARCore for AR plane tracking.
- Applied RANSAC plane fitting and prismatic integration methods to calculate volume from segmented depth data.
- Integrated the USDA FoodData Central API to convert estimated volume into nutritional information.
- Achieved a target accuracy of ≤15% MAPE (Mean Absolute Percentage Error) across 15 tested food items.
- Presented and defended findings before an academic thesis panel.

EDUCATION

Bachelor of Science in Computer Science — University of San Agustin, Iloilo City, Philippines*[2023] – [2027]*

- Thesis: AR-based mobile application for food volume estimation using monocular depth estimation
- [Add relevant coursework, honors, or GPA if applicable]

ADDITIONAL INFORMATION

- Graphic Design: Experience creating organizational charts, layouts, and visual assets using Figma and Canva
- Have minor experience in game development